Miles Woollacott

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PROFESSIONAL PROFILE

Fourth-year Ph.D. student in Statistics with a dual background in Statistics and Computer Science. Specializing in design and analysis of computer experiments and surrogate modeling for spatial processes. Experienced in Python and R.

EDUCATION

North Carolina State University, Raleigh, NC — Expected May 2028
Ph.D. in Statistics, Advisor: Dr. Jonathan Stallrich

North Carolina State University, Raleigh, NC — May 2026
Master of Research in Statistics, GPA: 3.50

University of Virginia, Charlottesville, VA — May 2023
Bachelor of Science in Statistics & Bachelor of Arts in Computer Science, Department GPA: 3.84, Cumulative GPA: 3.78

RESEARCH EXPERIENCE

Graduate Researcher | *North Carolina State University* | Raleigh, NC — August 2023-Present

- Develop sequential diagnostic methods for Gaussian Process surrogate models, creating time series-based frameworks for dynamic sample size determination for computer experiments.
- Create new designs for computer experiments based on filling the space of pairwise differences between points to improve hyperparameter estimation.
- Design a novel sensitivity analysis method within the NSF-funded UQ4Life Research Training Group to fine-tune hybrid AI/ML weather forecasting models, applying uncertainty quantification to improve short- and long-term predictive accuracy.

TEACHING EXPERIENCE

Seated Instructor | *North Carolina State University* | Raleigh, NC — August 2024-May 2026

- Taught introductory statistics to cohorts of 60+ undergraduate students by independently preparing and delivering comprehensive statistical lectures, course concepts, and support materials.
- Designed rigorous homework and exam questions, evaluated student performance, and provided constructive analytical feedback to improve academic outcomes.
- Held regular office hours to clarify concepts and boost student comprehension.

Hybrid Instructor | *North Carolina State University* | Raleigh, NC — August 2023-May 2024

- Led weekly recitation sessions to 3 sections of 40 undergraduate students each for introductory statistics by guiding students through in-class assignments, boosting understanding of course content and promoting student-instructor interaction.
- Designed rigorous exam questions, evaluated student performance, and provided constructive analytical feedback to improve academic outcomes.
- Held regular office hours to clarify concepts and boost student comprehension.

Teaching Assistant | *University of Virginia* | Charlottesville, VA — May 2021-June 2021, January 2023-May 2023

- Evaluated weekly R programming assignments for 400 students, providing detailed, actionable feedback on code efficiency, syntax, and visualization techniques.
- Held regular office hours to clarify concepts and boost student comprehension.

PREPRINTS

- Nafees, S., Pranathi Vemuri, V. N., **Woollacott, M.**, Solak, A. C., Logan, P., McGeever, A., ... & Rice, S. H. (2022). ortho_seqs: A Python tool for sequence analysis and higher order sequence–phenotype mapping. bioRxiv, 2022-09.

PRESENTATIONS

- **Woollacott, M.** & Stallrich, J. (2026). "Revisiting Sample Sizes for Sequential Computer Experiments." Oral presentation at the Fall Technical Conference, Charlotte, NC.
- **Woollacott, M.** & Stallrich, J. (2026). "Determining Sample Sizes for Sequential Computer Experiments." Poster presented at the Joint Statistical Meetings, Boston, MA.
- **Woollacott, M.** & Stallrich, J. (2026). "Determining Sample Sizes for Sequential Computer Experiments." Poster presented at the Quality and Productivity Research Conference, Santa Fe, NM.

INDUSTRY EXPERIENCE

Data Science (Automation) Intern | *IBM* | San Jose, CA May 2022-August 2022

- Improved system reliability and log anomaly detection accuracy by debugging and refactoring core Python automation modules for an internal process.
- Analyzed statistical relationships between complex log types using basket analysis in R to isolate patterns and discover downstream systemic issues.
- Collaborated within a Scrum methodology, delivering weekly metric-driven progress updates to multidisciplinary engineering stakeholders.

Data Science Intern | *Chan Zuckerberg BioHub* | San Francisco, CA June 2021-April 2022

- Streamlined data interpretation workflows for non-statistical research teams by engineering custom Python visualization tools to process tensor-based orthogonal polynomial outputs.
- Authored and maintained comprehensive software documentation, improving user comprehension of tool execution and analytical output.

HONORS AND AWARDS

- NSF Student Travel Award, *National Science Foundation*, August 2026
- NSF Student Travel Award, *National Science Foundation*, May 2026

PROJECTS

Merchandising Discounting Strategies and Impact on Revenue, Spring 2026
Completed as part of a Statistical Consulting course, in collaboration with the Carolina Hurricanes National Hockey League team. Engineered pricing elasticity models on 100,000+ transaction records using elastic net regression with autoregressive terms, isolating key promotional levers to optimize discounting strategy and revenue lift.

March Madness Predictor, Spring 2026
Completed as an independent predictive analytics project for the 2026 NCAA Men's Basketball Tournament. Scraped game-level ESPN data to engineer leave-one-out season metrics in Python, then fit a cross-validated elastic net logistic regression model with team random effects in R to rank competing teams. Correctly predicted 2 of the Final Four participants and accurately forecast several major first-round upsets.

A Primer on Projection Properties of Designs in Computer Experiments, Fall 2025
Completed for PhD requirements at North Carolina State University. Synthesized literature on projection-based computer experimental designs and executed a large-scale simulation study on NCSU High-Performance Computing (HPC) clusters. Evaluated variable selection using spike-and-slab priors on anisotropic lengthscale hyperparameters, demonstrating that standard designs match specialized projection designs and revealing critical tradeoffs between projection properties and classification stability.

Random Projection-Based Testing in High Dimensions, Fall 2024
Completed in a two-person collaboration for an Advanced Statistical Inference course. Evaluated the extension of a novel high-dimensional random projection test statistic (Liu & Huang, 2024) to Poisson distribution settings, running simulations across varying signal strengths and sparsity regimes. Confirmed that the novel test statistic maintains robust power and Type I error control under Poisson specifications, matching original benchmark literature.

LEADERSHIP EXPERIENCE

Cavalier Marching Band | *University of Virginia* | Charlottesville, VA

Drill Instructor August 2022-December 2022

- Collaborated with leadership to plan visual drill choreography for dynamic halftime shows performed by a 250+ member ensemble.
- Co-led bi-weekly, 2-hour rehearsals, instructing members on field execution, visual uniformity, and marching technique to elevate performance quality.
- Delivered individualized, real-time coaching to refine posture, alignment, and tempo execution across the section.

Webmaster August 2021-December 2021

- Designed and maintained an intuitive, user-friendly Excel system to monitor and track real-time attendance for over 250 band members across all rehearsals, game days, and special events.
- Streamlined administrative tracking, improving data accuracy and reporting efficiency for band administration.

Kappa Kappa Psi Honorary Band Fraternity | *University of Virginia* | Charlottesville, VA

Ritual Master August 2022-May 2023

- Led the planning and execution of official initiation rituals required for candidates transitioning into full brotherhood, ensuring adherence to national fraternity traditions and standards.
- Educated candidates and active members on organizational history, ethical standards, and ceremonial expectations.

Recording Secretary August 2021-May 2022

- Authored and distributed weekly chapter updates to ~20 active brothers to align team goals and communicate upcoming service events.
- Recorded formal meeting minutes and established an objective point-tracking system to monitor, evaluate, and encourage active member participation.
- Executed community outreach and logistical operations, including organizing instrument transport, leading music education visits to local elementary schools, and supporting local community initiatives.

ADDITIONAL SKILLS

Languages & Tools: Python (NumPy, Pandas, Scikit-Learn), R (Tidyverse, Parallel, JAGS), C++, Java, HPC, Git/GitHub, HTML, SQL
Statistics: Design of Experiments, Bayesian Statistics, Spatial Statistics, Sports Statistics, Machine Learning
Technical: Overleaf, LaTeX
Languages: English-Fluent, French-Elementary